Weekly Progress Report 1

1. Cover Information

Project title: GarmentIQ
Team name: The Juls
Project Description: Data-driven Appeal Operations & Sales Intelligence System
Team Leader: Garv Dudy
Date: Friday, March 13th, 2026

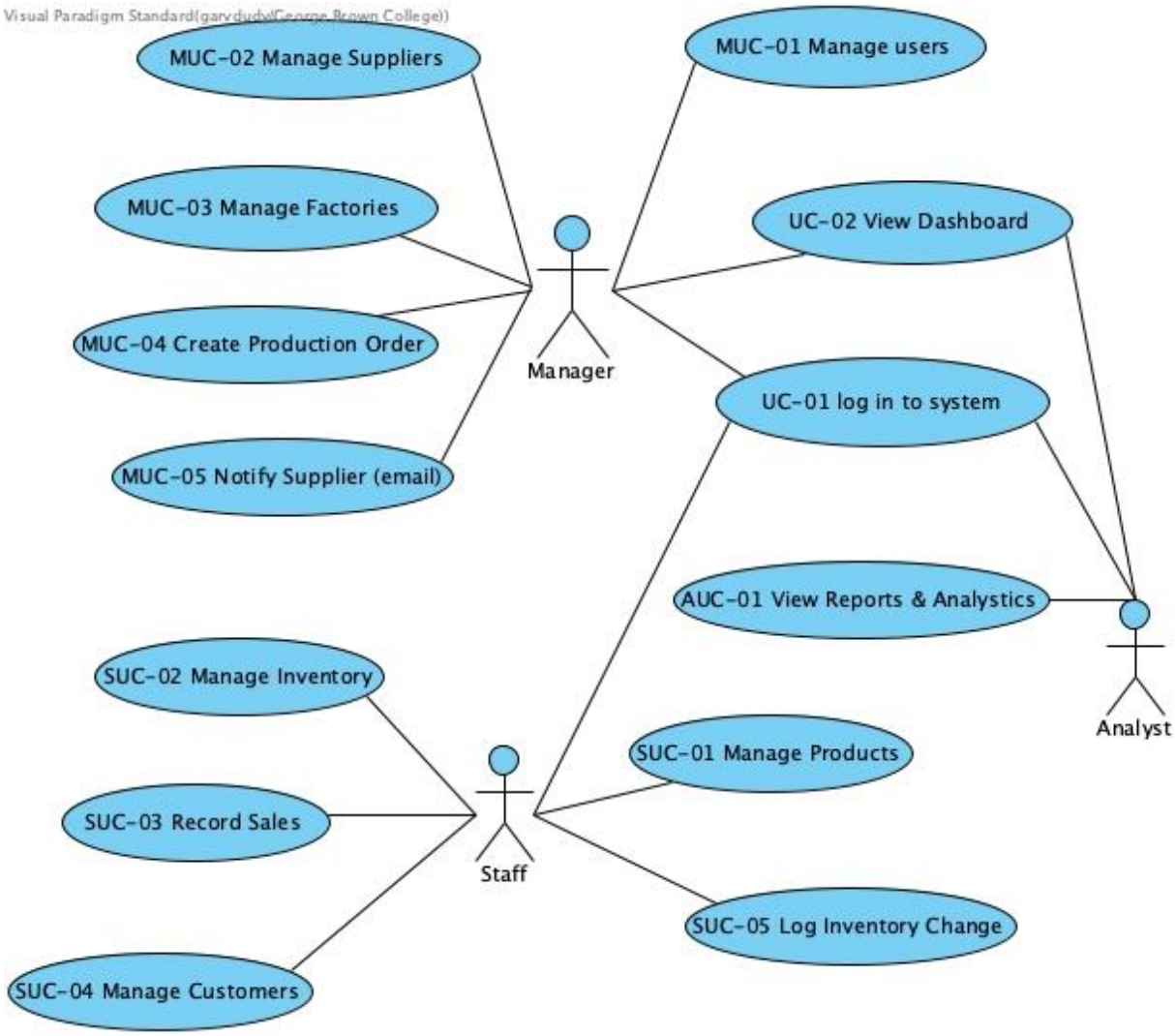
2. Summary of This Week

During this reporting period, the focus of the project was on system analysis, design, and database preparation. The goal was to clearly define the system structure and data relationships before starting implementation.

Several system modeling techniques were used to understand the system workflow and interactions between users, suppliers, and factories. Diagrams were created to represent system functionality, process flow, and database structure. After completing the design phase, a dataset was created and imported into SQL to prepare the system for further development and analysis.

3. Milestones and Completed Work

1. **Use Case Diagram Creation**
   Created system use case diagrams to identify user interactions and system functionality using Visual Paradigm.



2. **Textual Analysis Diagrams**

   Documented system behavior and functional requirements using textual analysis models.

Textual Analysis – GarmentIQ Case Study
Scenario: Manager logs in, reviews operations, and notifies supplier

UC-01 – Log in to system Sarah Mitchell (Manager) opens the GarmentIQ web portal and enters her credentials (sarah.mitchell@gmail.com, username garment-iq-sarah1991). The system validates her identity against the USERS table, confirms her role_id = 1 (Manager) from the ROLES table, and grants her full access to the dashboard and all management modules.

UC-02 – View dashboard Upon login, Sarah lands on the manager dashboard. The system queries the SALES, PRODUCTS, and INVENTORY_LOG tables to display key metrics: total revenue this week, top-selling products, and current low-stock alerts. She notices that the "Dark Blue Hoodie (Size M)" has fallen below its reorder_level of 25 units.

MUC-01 – Manage suppliers Sarah navigates to the Suppliers module. The system retrieves all records from the SUPPLIERS table. She identifies that the responsible supplier for hoodies is *Sakura Textiles Co.* (supplier_id = 1, Japan). She reviews their contact details – kenji.yamamoto@sakuratextiles.jp – and notes the last

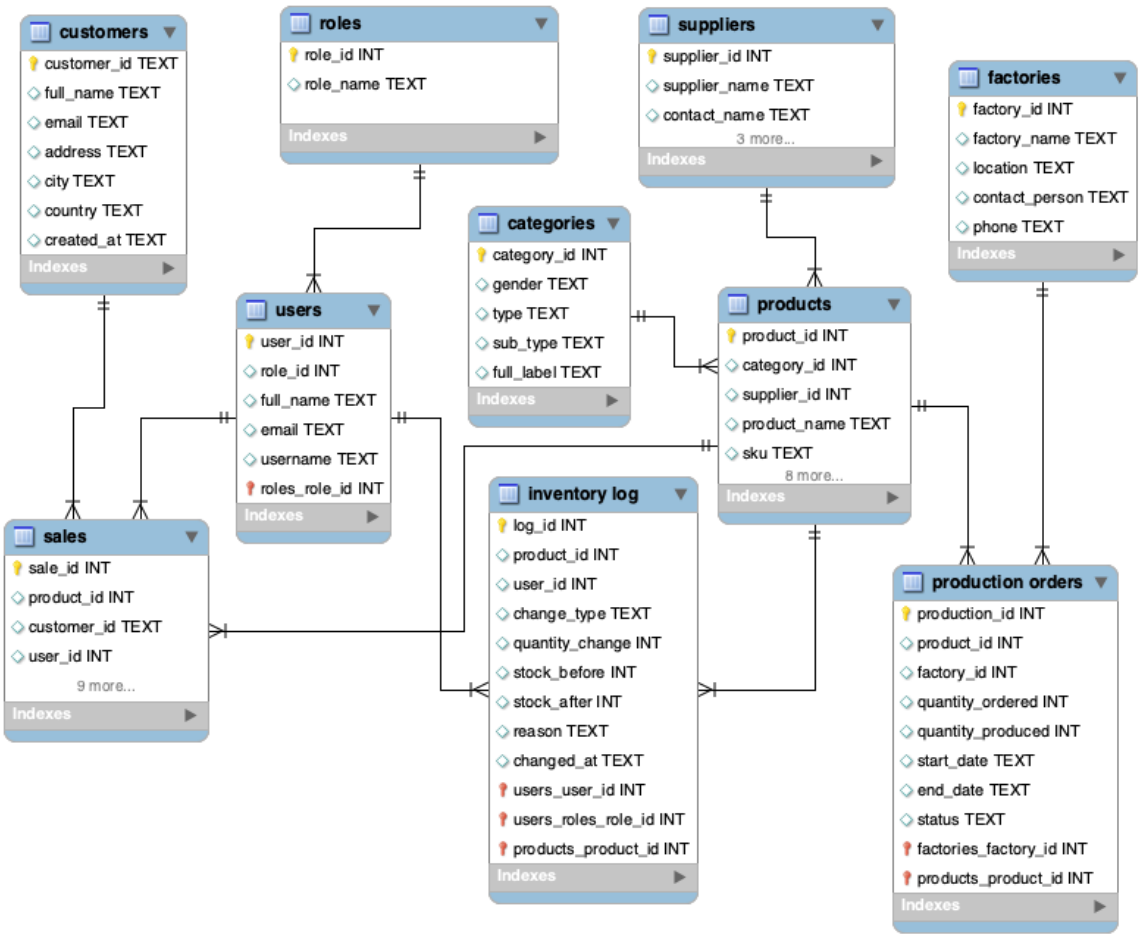
No.	Candidate Class	Extracted Text	Type	Description	Occurrence	Highli...
1	Sarah The Ma	Sarah	Actor	Sarah Mitchell is the Manager if GarmentIQ and she has access to many different thinks which includes	6	
2	Supplier Table	SUPPLIERS tabl	Use Case	Table Where Datials of All suppliers connected to company is present	1	
3	Factorie Table	FACTORIES tab	Use Case	Table where Details Related to Factories where the products been manafactured is stored along with cor	1	
4	Staff	Staff	Class	Staff	2	
5	Manager	Manager	Class	Manager	4	
6	Analyst	Analyst	Class	Analyst	2	
7	John The Sta	John	Actor	John Doe is the Staff Mamber he can manage inventery, recode the sales , manage customers and can p	5	
8	James The An	James	Actor	James is the lead business and Data analyst of the company and is taksed to cleaned the provided data	1	
9	Returning Cus	returning custo	Use Case	This is the big and best as well as most important key of the company	1	
10	Customer Tab	CUSTOMERS ta	Use Case	Table Whwere Details of all cusotmers is stored along with name, shipping address, email, cstm_id and	1	

3. **Class Diagram Design**
   Designed system entities and relationships using class diagrams in Visual Paradigm.
4. **Database Table Planning**
   Converted the class diagram structure into 10 database tables representing system entities such as users, roles, suppliers, factories, and customers.
5. **Dataset Preparation**
   Created an Excel dataset with dummy data including 500 customer records for testing and analysis purposes using Microsoft Excel.

	A	B	C	D	E	F	G
1	customer_id	full_name	email	address	city	country	created_at
2	cstm26-0001	Chloe Flores	chloe.flores@gmail.c	957 Maple Dr	New York	USA	2023-09-07 00:00:00
3	cstm26-0002	Joseph Roberts	joseph.roberts@gmai	510 Oak Ave	New York	USA	2022-11-22 00:00:00
4	cstm26-0003	Logan Carter	logan.carter@gmail.c	793 Queen Rd	Berlin	Germany	2024-10-08 00:00:00
5	cstm26-0004	Maria Taylor	maria.taylor@gmail.c	888 Main St	Chicago	USA	2024-06-30 00:00:00
6	cstm26-0005	Jack Thomas	jack.thomas@gmail.c	399 Main St	Berlin	Germany	2024-12-14 00:00:00
7	cstm26-0006	Henry Torres	henry.torres@gmail.c	93 King Blvd	São Paulo	Brazil	2022-11-27 00:00:00
8	cstm26-0007	Joseph Garcia	joseph.garcia@gmail.	415 Oak Ave	São Paulo	Brazil	2022-07-28 00:00:00
9	cstm26-0008	Matthew Hill	matthew.hill@gmail.c	699 Main St	Los Angeles	USA	2024-11-03 00:00:00
10	cstm26-0009	Michael Allen	michael.allen@gmail.	450 King Blvd	Vancouver	Canada	2024-03-12 00:00:00
11	cstm26-0010	Charlotte Turner	charlotte.turner@gm	836 King Blvd	Berlin	Germany	2023-11-22 00:00:00
12	cstm26-0011	Victoria White	victoria.white@gmail	501 King Blvd	Chicago	USA	2022-12-25 00:00:00
13	cstm26-0012	Evelyn Thomas	evelyn.thomas@gma	844 Main St	Houston	USA	2024-07-23 00:00:00
14	cstm26-0013	Penelope Martin	penelope.martin@gm	248 Main St	Chicago	USA	2023-02-24 00:00:00
15	cstm26-0014	Chloe Moore	chloe.moore@gmail.c	387 Queen Rd	Paris	France	2024-02-14 00:00:00
16	cstm26-0015	Samuel Hill	samuel.hill@gmail.co	273 Main St	Chicago	USA	2022-03-23 00:00:00
17	cstm26-0016	Ethan Johnson	ethan.johnson@gmai	271 Queen Rd	Chicago	USA	2022-12-06 00:00:00
18	cstm26-0017	Elijah Brown	elijah.brown@gmail.c	842 Maple Dr	Los Angeles	USA	2022-11-11 00:00:00
19	cstm26-0018	Chloe Torres	chloe.torres@gmail.c	818 Queen Rd	Los Angeles	USA	2023-06-23 00:00:00
20	cstm26-0019	Noah Hall	noah.hall@gmail.com	475 King Blvd	Los Angeles	USA	2023-10-21 00:00:00
21	cstm26-0020	Noah Brown	noah.brown@gmail.c	747 Maple Dr	Mumbai	India	2024-07-12 00:00:00
22	cstm26-0021	Nora Green	nora.green@gmail.cc	152 Queen Rd	Chicago	USA	2023-02-20 00:00:00
23	cstm26-0022	Zoey Hall	zoey.hall@gmail.com	892 Maple Dr	Vancouver	Canada	2022-01-24 00:00:00
24	cstm26-0023	Sebastian Adams	sebastian.adams@gn	858 Queen Rd	Seoul	South Korea	2024-09-09 00:00:00
25	cstm26-0024	Emily Johnson	emily.johnson@gmail	691 Maple Dr	Mumbai	India	2024-11-03 00:00:00
26	cstm26-0025	Aiden Johnson	aiden.johnson@gmail	691 Maple Dr	Los Angeles	USA	2024-03-15 00:00:00

ROLES | USERS | CUSTOMERS | CATEGORIES | SUPPLIERS | PRODUCTS | FACTORIES | PRODUCTION_ORDERS | SALES | INVENTORY_LOG

6. **SQL Data Import and Analysis Preparation**
   Imported the prepared dataset into SQL for database analysis, reverse engineering, and EER diagram generation.



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4. Work in Progress

1. **Process Flow / UML Workflow Diagram**: Creating workflow diagrams including start and end processes to visualize system operation using draw.io.
2. **Preparing application structure**: planning UI and backend modules before development begins.

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5. Blockers, Issues, and Risks

Currently there are no major blockers affecting project progress.

Potential risk:
Complex database relationships may require additional adjustments during implementation & Issues might occur while connecting frontend to backend and writing backend logics.

Mitigation:
Performing early schema validation and EER review to ensure correct relationships between entities.

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6. Plan for Next Week

1. Finishing Process Flow for best backend logics.
2. Begin UI design and development of the web application.
3. Develop login page and authentication system.
4. Start implementing backend logic and pgAdmin4 database connection.
5. Working with complex quarries in MySQL.

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7. Architecture, Design, and Technical Decisions

The system design follows a structured layered approach consisting of:

- Frontend layer: Web interface for Customers as well as Staff for system interaction.
- Application layer: Backend logic handling system operations.
- Database layer: SQL/pgAdmin4 database storing system data.

System modeling and architecture diagrams were created using Visual Paradigm and draw.io to clearly define relationships between system components.

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8. Implementation, Testing, and Quality Updates

At this stage, the project is still in the design and database preparation phase. Implementation of the web application and backend logic will begin in the next development cycle.

Initial datasets and database structures have been prepared to support upcoming development and testing activities.

9. Project Management and Collaboration Updates

Project tasks and development progress are being tracked using version control and task management tools. while future project data tracking may also involve Alteryx for data automation and analysis whereas Jira is being in use more than any other platform for monitoring development tasks and progress. All the recourses of project are available on **GitHub Repository:** **https://github.com/Garvdudy/GarmentIQ** and code versions will be maintained through GitHub and Vercel.

10. Reflections and Lessons Learned

This phase helped establish a clear understanding of the system requirements and architecture. Creating diagrams before implementation allowed better visualization of system functionality and data flow. Preparing structured datasets early also ensures smoother testing and data analysis during later development stages.

11. Attachments and References

- Use Case Diagram
- Textual Analysis Diagram
- Excel Dataset with Sample Data
- SQL database structure and EER diagram
- Link to GitHub with all Up-to-date Resources